GCCANet: A Small-Object Sensitive and Environment-Robust Network for Remote Sensing Change Detection

Jingqi Chen, Yonghong Song, *Member, IEEE*, Yifei Gao, You Su, and Yong Yu

***Abstract*—Remote sensing change detection (RSCD), widely used in industrial and military applications, aims to accurately identify surface changes by comparing two temporally separated images of the same region. In recent years, deep learning has greatly advanced the development of RSCD. However, in complex environments, subtle and background-induced changes often compromise the accuracy of change detection (CD). To address these challenges, we propose the global-context and change-aware network (GCCANet). We adopt a dual-encoder architecture to extract rich semantic features and design an adaptive layer attention (ALA) module that condenses multilevel Transformer semantics into compact global guidance. We further design a background suppression module (BSM) to inject this global guidance into each convolutional neural network (CNN) stage, effectively suppressing background noise and stabilizing the subsequent differencing process. Moreover, we design a difference learning module (DLM) that couples local differencing with global attention matching to extract robust and detail-aware change cues. Then, we introduce a lightweight decoder equipped with a multiscale small-object enhancement (MS-SOE) strategy to enhance subtle changes and preserve thin structures. Extensive experiments on five public benchmark datasets demonstrate that GCCANet consistently achieves state-of-the-art (SOTA) performance. The code is available at https://github.com/E1ison/GCCANet**

***Index Terms*—Change detection (CD), deep learning, remote sensing, small-object CD.**

I. Introduction

REMOTE sensing change detection (CD) is the process of identifying and quantifying changes in an object, phenomenon, or landscape by comparing images acquired through remote sensing technology at different times, and thereby supporting a broad range of critical applications in urban planning [1], disaster response [2], urban management [3], and environmental monitoring [4], [5]. Small-object changes are characterized by limited scale, weak texture, and low contrast, making them easily obscured by complex backgrounds or illumination artifacts. These changes often correlate with practical applications, such as detecting new small buildings, local facility modifications, or minor environmental perturbations. Accurately identifying these subtle variations thus remains a critical and challenging task in remote sensing CD [6].

In the early stages of CD research, most methods relied on handcrafted features and can be broadly categorized into pixel-based and object-based approaches [7]. Pixel-based methods compare bi-temporal pixel values using algebraic operations, transform-based analysis, or traditional classifiers [8], [9], [10], whereas object-based methods aggregate pixels into coherent objects to enable more flexible region-level comparisons and segmentation strategies [11]. Despite their simplicity, these handcrafted approaches exhibit inherent limitations: pixel-based methods are sensitive to noise and lack spatial contextual modeling, while object-based methods suffer from poor cross-scene stability, making consistent and reliable performance difficult to achieve.

The emergence of deep learning has led CD methods to converge toward an end-to-end paradigm that encodes bi-temporal images, performs feature-level interaction or differencing, and decodes a dense change map (CMap). Within this paradigm, convolutional neural network (CNN)-based methods have long been dominant, especially since the introduction of fully convolutional networks (FCNs) for CD by Mou et al. [13]. Many representative approaches follow a Siamese or dual-branch design and process bi-temporal images in parallel to extract and compare multilevel features [14], [15]. Although these methods are effective [16], they are constrained by limited receptive fields and their reliance on local differencing, which makes them susceptible to pseudo-changes induced by illumination and seasonal variations. Recent studies have introduced attention mechanisms [17], [18] and Transformers [19] to enhance detection capability and overcome the locality limitations of CNNs [20]. However, Transformers that only learn global information from the training set are not sufficiently robust, and existing architectures still lack effective mechanisms for integrating complementary information from CNN and branches. Moreover, bi-temporal feature fusion remains challenging, as simple methods like subtraction are highly sensitive to illumination and often produce noisy predictions with blurred boundaries. Furthermore, critical fine details are lost during decoding, as shown in Fig. 1. Visual backbones can extract rich multilevel features, including fine-grained cues from small objects, but many of these details are lost during the upsampling process, which is especially

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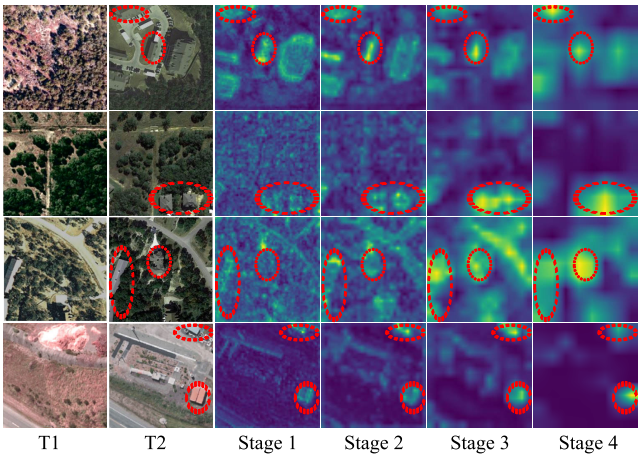


Fig. 1. Visualization of multistage differential features extracted by visual backbones.

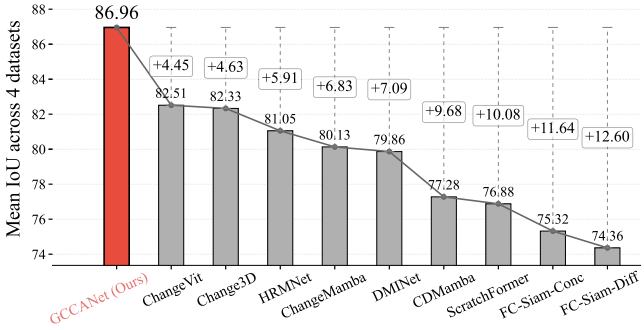


Fig. 2. Performance improvements of GCCANet over previous methods on four benchmark datasets.

harmful for small-scale changes and leads to many missed detections in existing methods. The performance of GCCANet is shown in Fig. 2, achieving state-of-the-art (SOTA) levels.

To this end, we propose the global-context and change-aware network (GCCANet). When combined with CNNs and powerful large-scale self-supervised backbones such as DINO [21], these models provide robust and general-purpose semantic representations [22], [23]. Motivated by this, our GCCANet leverages a dual-encoder architecture to extract rich multilevel features and integrates several novel components to address the aforementioned challenges. First, we introduce the adaptive layer attention (ALA) module and the background suppression module (BSM) to distill expert-level global guidance from the Transformer features and use this guidance to suppress background noise in the CNN features, thereby achieving CNN and Transformer interaction. Furthermore, our difference learning module (DLM) robustly models temporal relationships to extract detail-aware change cues. Finally, our multiscale small-object enhancement (MS-SOE) strategy (visualized in Fig. 3), implemented in a lightweight deeply supervised decoder (DSD), ensures the effective refinement of the CMap and the reliable enhancement of subtle cues.

In summary, the main contributions are as follows.

1) The ALA module is designed to aggregate multilevel features from the Transformer branch, forming a compact and expert-level global guidance feature.
2) BSM leverages the expert guidance generated by ALA, fusing it with local CNN features. This process explicitly models and filters background noise to suppress pseudo-changes and stabilize downstream difference learning.
3) DLM unifies lightweight local differencing, using a tri-branch input, with attention-based global matching to effectively yield boundary-aware change cues that remain robust to illumination and seasonal variance.
4) We introduce a lightweight small-object prediction head, coupled with a deep supervision strategy, to enhance the recall and delineation of subtle, small-scale changes.
5) GCCANet achieves superior performance across five benchmark datasets and shows strong robustness in both semi-supervised and cross-domain CD.

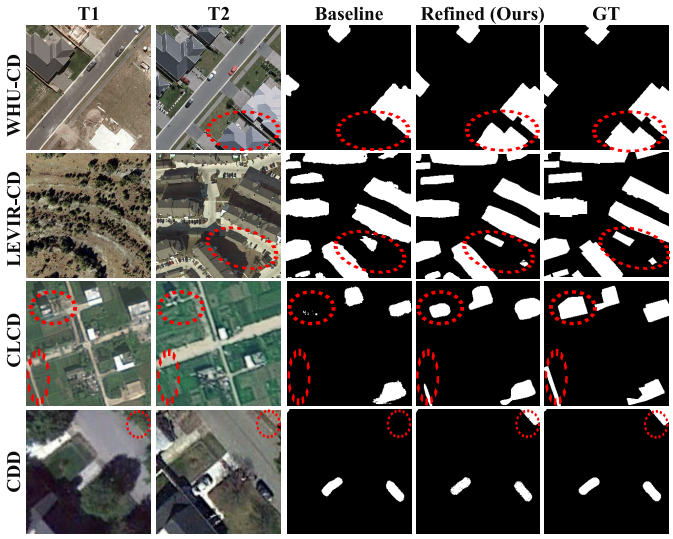


Fig. 3. Effect of the proposed MS-SOE. T1 and T2 denote the prechange and postchange images. The baseline column shows the prediction results from ScratchFormer [12] for comparison. MS-SOE helps detect small and medium-sized targets while reducing missed detections on all four datasets.

II. Related Work

A. Deep Learning in CD

Deep learning for bi-temporal CD has evolved from early Siamese CNN pipelines toward attention-centric and state-space formulations. Within CNN-based research, Alcantarilla et al. [24] demonstrated pixel-wise structural CD in street-view videos using FCNs, validating end-to-end dense prediction in real scenes. Shao et al.'s SUNet [25] introduced a dual-channel design augmented with edge cues to handle heterogeneous satellite, unmanned aerial vehicle (UAV) imagery, and SNUNet-CD [16] employed densely connected Siamese Nested-UNet to better preserve shallow details and boundaries for small changes. Building on the need for long-range cross-time reasoning, Transformer approaches became prominent: the BIT [20] framework formulates CD as token-based representation learning to capture spatial–temporal relations efficiently. ChangeFormer [26] adopts a pure Transformer Siamese encoder with a multilayer perceptron (MLP) decoder for multiscale feature recovery. More recently, state-space models (SSMs), including earlier structured variants such as S4 [27] and S5 [28], as well as selective Mamba-style architectures, have provided efficient paradigms for long-range

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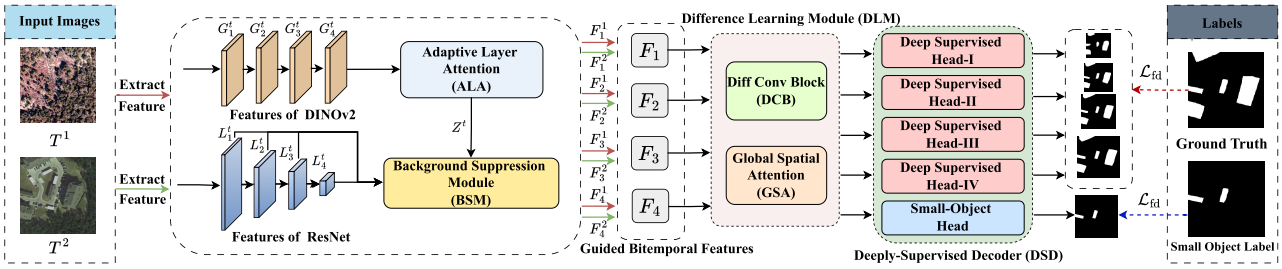


Fig. 4. Framework of the proposed GCCANet. GCCANet employs DINOv2 and ResNet to extract multilevel bi-temporal features. The ALA module reweights the features from DINOv2 to generate an expert feature, which is then fused with the ResNet-extracted features through BSM, producing four fused feature maps. These fused bi-temporal features are subsequently processed by DLM to obtain discriminative change features. DSD refines these features to produce the final predictions. Training employs deep supervision and a dedicated small-object head to enhance multiscale features and subtle changes.

modeling with high-resolution inputs, while existing remote sensing CD (RSCD) studies mainly concentrate on Mamba-derived designs. For example, ChangeMamba [29] introduces spatio-temporal SSM mechanisms across binary, semantic, and damage assessment settings, whereas RS-Mamba [30] addresses dense prediction on very-high-resolution imagery without patching through omnidirectional selective scans. Meanwhile, spiking neural networks (SNNs) have emerged as a promising paradigm for spatio-temporal representation learning. By encoding information with discrete spikes over time, SNNs naturally support event-driven computation and efficient temporal modeling, making them well-suited for dynamic vision and resource-constrained scenarios. Recent studies have explored SNN-based approaches in remote sensing and related spatio-temporal visual tasks [31], [32], [33], highlighting their potential for CD that relies on modeling temporal variations across multitemporal observations. However, many methods ignore prior knowledge and local feature interactions in bi-temporal fusion, thus remaining highly sensitive to illumination and seasonal variations.

B. Small-Object CD in Remote Sensing

Detecting small-scale and subtle changes is critical in RSCD, as these fine-grained variations often indicate minor land-use transitions, localized infrastructure updates, or early risk signals of practical importance. Seo et al. [34] proposed a feature-based approach for small-object CD from high-resolution satellite imagery, which uses feature points rather than pixels or object segments as change units. MAEANet [35] leverages multiscale attention with edge-aware priors to improve the visibility and separability of fine changes on high-resolution data; STNet [36] jointly models temporal gating and cross-scale attention to reduce semantic ambiguity and enhance fine-grained change perception across benchmarks. Deep supervision and attention are further used to improve recall in small or low-contrast regions, as illustrated by DSAMNet [37], which mitigates small-object dilution during multilevel fusion. On the data side, research is moving toward small and rare changes, and the china land cover dataset (CLCD) [38] cropland-change dataset provides numerous subtle conversion samples that support targeted evaluation. However, existing methods do not fully attend to small-scale change regions and often lose small-object information during decoding, leading to missed detections.

III. Methodology

A. Framework

As illustrated in Fig. 4, the proposed GCCANet employs a dual-stream backbone that extracts four-scale bi-temporal features. For the CNN branch, we adopt the residual network (ResNet) [39] architecture to leverage its hierarchical features. For the Transformer branch, we adopt DINOv2 [21] to capture semantic and global contextual information. Semantic features refer to object-level representations that encode structural patterns and long-range spatial dependencies across the scene, corresponding to semantically meaningful regions such as buildings or road structures in remote sensing imagery. Such representations emphasize category-consistent structures and large-scale spatial organization, which are more robust to illumination variations and seasonal changes. The input consists of a pair of bi-temporal images $\mathbf{T}^t \in \mathbb{R}^{H\times W\times 3}$, $t \in \{1, 2\}$. For each time step, the input image goes through a dual-stream backbone, yielding two pyramids of multiscale features. The Transformer branch (DINOv2) captures global semantic context $\mathbf{G}_k^t$, while the CNN branch (ResNet) extracts local structural cues $\mathbf{L}_k^t$, including edges, textures, and fine spatial details that are essential for detecting small objects and preserving boundary information. Subsequently, an ALA module processes the DINOv2 pyramid to form a compact guidance feature $\mathbf{Z}^t$. This guidance is then fused into the local multiscale feature pyramid via the BSM to obtain guided bi-temporal features $\mathbf{F}_k^t$, $k \in \{1, 2, 3, 4\}$. Following that, the DLM refines these features to learn discriminative change features $\mathbf{Y}_k$, $k \in \{1, 2, 3, 4\}$. Finally, the DSD processes these features to produce the main CMap $\hat{\mathbf{y}}_m$, three auxiliary stage-wise CMaps $\hat{\mathbf{y}}_k$, $k \in \{2, 3, 4\}$, and a small-object map $\hat{\mathbf{y}}_s$. Training optimizes a composite objective $\mathcal{L}$ based on the $\mathcal{L}_{\text{fd}}$ loss (a combination of Focal and Dice loss) applied to the main map, the auxiliary side heads, and the small-object branch. After convergence, the main CMap $\hat{\mathbf{y}}_m$ serves as the final predicted CMap.

B. Adaptive Layer Attention

Transformer features from DINOv2 provide high-level semantic representations encoding object structures and long-range spatial dependencies, but their semantic granularity varies across stages and may contain noise. ALA learns the importance of each level and re-aggregates the pyramid at a unified resolution to produce an input-adaptive guidance feature. As shown in Fig. 5, each stage feature $\mathbf{G}_k^t$ is first

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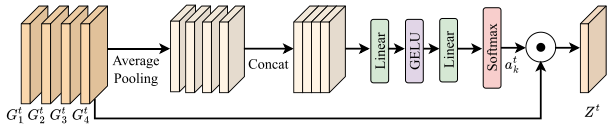


Fig. 5. Structure of the ALA module. The ALA dynamically computes weights to fuse the multistage DINOv2 features into a single guidance feature for each time step.

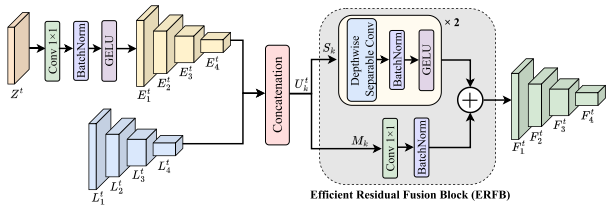


Fig. 6. Structure of BSM. The ALA guidance is aligned to each CNN stage and fused via an ERFB, injecting global priors into local features.

processed by a global average pooling (GAP) layer to generate a channel-wise descriptor $\mathbf{z}_k^t$

$$\mathbf{z}_k^t = \mathrm{GAP}\left(\mathbf{G_k^t}\right). \tag{1}$$

The four descriptors are concatenated and fed to a two-layer MLP

$$\mathbf{v}^t = \left[\mathbf{z}_1^t, \mathbf{z}_2^t, \mathbf{z}_3^t, \mathbf{z}_4^t\right] \tag{2}$$

$$\mathbf{h}^t = \phi\left(W_1\mathbf{v}^t + \mathbf{b}_1\right) \tag{3}$$

$$\mathbf{q}^t = W_2\mathbf{h}^t + \mathbf{b}_2 \tag{4}$$

where $\mathbf{v}^t$ denotes the concatenated feature vector formed by stacking the four stage-wise descriptors $\mathbf{z}_k^t$. The first linear transformation projects $\mathbf{v}^t$ into a bottleneck representation $\mathbf{h}^t$ with Gaussian error linear unit (GELU) activation ϕ, and the second linear layer outputs four attention scores $\mathbf{q}^t$. W_1 and W_2 are learnable weight matrices, while $\mathbf{b}_1$ and $\mathbf{b}_2$ are bias terms. A softmax operation on $\mathbf{q}^t$ yields the normalized attention weights $\boldsymbol{a}_k^t$, which adaptively reflect the importance of each feature level k

$$\boldsymbol{a}_k^t = \mathrm{Softmax}\left(\mathbf{q}^t\right). \tag{5}$$

Finally, the guidance feature for time step t is formed as a convex combination of the pyramid levels

$$\mathbf{Z}^t = \sum_{k=1}^{4} \boldsymbol{a}_k^t \mathbf{G}_k^t. \tag{6}$$

C. Background Suppression Module

Pseudo-changes caused by illumination and seasonal variations often contaminate local features and lead to false positives (FPs), since local texture variations are difficult to distinguish from real structural changes when relying solely on CNN features. Therefore, BSM introduces the semantic guidance $\mathbf{Z}^t$ as a structural prior to refine the multilevel ResNet features $\mathbf{L}_k^t$. By injecting the guidance feature into each CNN stage, the network can condition local responses on the global semantic context, encouraging activations that are consistent with scene structures while suppressing responses caused by background appearance variations. In particular, semantic representations of unchanged background regions tend to remain consistent across the two timestamps, which

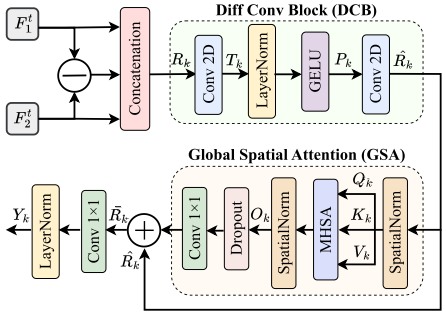


Fig. 7. Structure of DLM. BSM-injected bi-temporal features are concatenated as pre, post, and absolute difference, processed by a DCB for local cues, and then refined by GSA for long-range consistency.

encourages the fused features to become more similar and reduces feature differences caused by illumination or texture variations. In contrast, when real structural changes occur, the semantic responses differ, and the fused features remain discriminative, allowing the change signal to be preserved. As shown in Fig. 6, the injection process begins by aligning the global guidance $\mathbf{Z}^t$ to each local feature $\mathbf{L}_k^t$. For each stage $k \in \{1, 2, 3, 4\}$, where $\mathbf{L}_k^t$ has a spatial size of $H_k \times W_k$ and C_k channels, the guidance $\mathbf{Z}^t$ is first resized to (H_k, W_k) and then projected by a convolution-normalization mapping ψ_k, before being concatenated with $\mathbf{L}_k^t$

$$\mathbf{E}_k^t = \psi_k\left(\mathrm{Resize}_{H_k,W_k}\left(\mathbf{Z}^t\right)\right) \tag{7}$$

$$\mathbf{U}_k^t = \mathrm{Concat}_C\left(\mathbf{L}_k^t, \mathbf{E}_k^t\right) \tag{8}$$

where $\mathbf{E}_k^t$ denotes the adapted guidance feature, obtained by resizing and linearly projecting $\mathbf{Z}^t$ to match the spatial and channel dimensions of the kth CNN stage, and ψ_k represents a convolution followed by normalization and activation. The resulting tensor $\mathbf{U}_k^t$ integrates local texture information with global semantic priors, enabling subsequent convolutions to adaptively reweight local responses and suppress background noise.

To fuse the local and global cues within $\mathbf{U}_k^t$, we employ an efficient residual fusion block (ERFB) composed of two paths. The first path preserves spatial details and aligns channels using a 1×1 convolution with batch normalization, while the second enhances local context through depthwise separable convolutions

$$\mathbf{S}_k^t = \mathrm{BN}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{U}_k^t\right)\right) \tag{9}$$

$$\mathbf{M}_k^t = \mathrm{BN}\left(\mathrm{DWConv}_{3\times3}\left(\mathrm{GELU}\left(\mathrm{DWConv}_{3\times3}\left(\mathbf{U}_k^t\right)\right)\right)\right) \tag{10}$$

$$\mathbf{F}_k^t = \mathrm{GELU}\left(\mathbf{S}_k^t + \mathbf{M}_k^t\right) \tag{11}$$

where $\mathbf{S}_k^t$ serves as the shortcut branch that retains spatial details, and $\mathbf{M}_k^t$ is the residual branch that captures local cues through two depthwise convolutions. Their outputs are summed and activated by GELU to produce the fused feature $\mathbf{F}_k^t$.

D. Difference Learning Module

The DLM aims to separate genuine bi-temporal changes from background variability and thereby improve overall change reliability. As shown in Fig. 7, it operates on the BSM-injected preimage ($\mathbf{F}_k^1$) and postimage ($\mathbf{F}_k^2$) features at each level k. Many existing methods rely on simple feature differencing ($|\mathbf{F}_k^1 - \mathbf{F}_k^2|$) [25], [36], which provides limited context

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and discards the original state information. To capture richer contextual cues, a tri-branch input is therefore constructed by channel-wise concatenation

$$\mathbf{R}_k = \mathrm{Concat}_C\left[\mathbf{F}_k^1, \mathbf{F}_k^2, \left|\mathbf{F}_k^1 - \mathbf{F}_k^2\right|\right]. \tag{12}$$

Local difference modeling on this tri-branch input $\mathbf{R}_k$ is performed by the difference convolution block (DCB), which refines local cues through a sequence of convolutions and normalization operations to produce more stable representations. The computation process is summarized as follows:

$$\mathbf{T}_k = \mathrm{Conv}_{3\times3}\left(\mathbf{R}_k\right) \tag{13}$$

$$\mathbf{P}_k = \mathrm{GELU}\left(\mathrm{LN}\left(\mathbf{T}_k\right)\right) \tag{14}$$

$$\hat{\mathbf{R}}_k = \mathrm{Conv}_{3\times3}\left(\mathbf{P}_k\right). \tag{15}$$

The first 3×3 convolution compresses channels and mixes local features while maintaining spatial resolution and computational efficiency. Layer normalization followed by GELU activation stabilizes the feature distribution and enhances nonlinear representations, which have been empirically shown to preserve fine structural details [40]. The second 3×3 convolution aggregates neighboring information into compact local-difference representations, preparing the features for subsequent attention-based refinement.

While the DCB captures local cues, it lacks a mechanism for modeling long-range spatial dependencies. To address this at a controlled computational cost, the resulting feature $\hat{\mathbf{R}}_k$ is refined by a global spatial attention (GSA) block

$$[\mathbf{Q}_k, \mathbf{K}_k, \mathbf{V}_k] = \mathrm{Conv}_{1\times1}\left(\mathrm{LN}\left(\hat{\mathbf{R}}_k\right)\right) \tag{16}$$

$$\mathbf{O}_k = \mathrm{MHSA}\left(\mathbf{Q}_k, \mathbf{K}_k, \mathbf{V}_k\right) \tag{17}$$

where the input feature $\hat{\mathbf{R}}_k$ is first normalized and linearly projected through a shared 1×1 convolution to generate the query, key, and value tensors, stabilizing the subsequent attention computation and preserving spatial correspondence. Multihead self-attention (MHSA) is then applied to aggregate global contextual information across the entire feature map. Finally, the output is restored to the internal width and fused with the input through a residual connection

$$\bar{\mathbf{R}}_k = \hat{\mathbf{R}}_k + \mathrm{Dropout}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{O}_k\right)\right) \tag{18}$$

where dropout regularizes the global pathway. The final stage of the DLM performs output feature alignment via layer normalization and a 1×1 convolution

$$\mathbf{Y}_k = \mathrm{Conv}_{1\times1}\left(\mathrm{LN}\left(\bar{\mathbf{R}}_k\right)\right) \tag{19}$$

so that each scale is projected to a shared width and can be uniformly fused and supervised. The DLM thus yields a hierarchy of refined, scale-wise difference features $\mathbf{Y}_k$, $k \in \{1, 2, 3, 4\}$ prepared for subsequent fusion and processing.

E. DSD and Optimization Strategy

As shown in Fig. 8, DSD aggregates stage-wise change embeddings from coarse to fine to preserve contextual cues. $\mathbf{Y}_k$ denote the DLM outputs at $k \in \{1, 2, 3, 4\}$. Features are progressively refined by shallow residual blocks and a top-down pathway, where Res is a residual block with a 3×3 convolution, normalization, GELU, and a skip connection, and Up is a transposed convolution that doubles spatial resolution

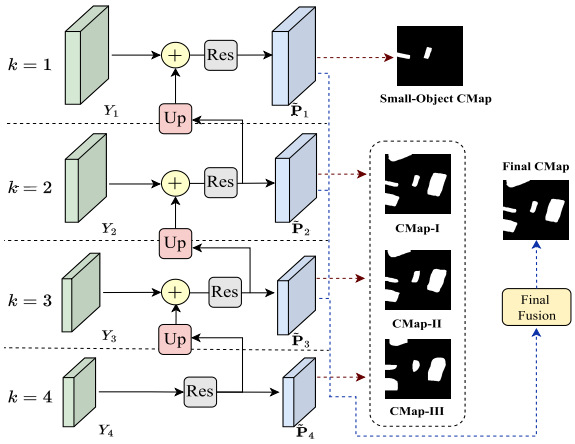


Fig. 8. Structure of DSD. Stage-wise embeddings are progressively refined along a coarse-to-fine top-down path using Up and Res blocks. Intermediate CMaps (CMap-I/II/III, corresponding to $\hat{\mathbf{y}}_k$) are generated by side heads at the coarser levels, while all features are fused at the finest level to produce the final CMap ($\hat{\mathbf{y}}_m$). A dedicated small-object head is also attached to this finest-resolution stage to output an auxiliary small-object CMap ($\hat{\mathbf{y}}_s$).

$$\tilde{\mathbf{P}}_k = \begin{cases} \mathrm{Res}\left(\mathbf{Y}_4\right), & \text{if } k = 4 \\ \mathrm{Res}\left(\mathbf{Y}_k + \mathrm{Up}\left(\tilde{\mathbf{P}}_{k+1}\right)\right), & \text{if } k \in \{1, 2, 3\}. \end{cases} \tag{20}$$

Algorithm 1 Pseudocode of Online SO-GT Generation

Input: Training batch GT $\mathbf{y}_{batch} = \{\mathbf{y}_b\}_{b=1}^{B}$
Output: Small-object GT batch $\mathbf{y}_s$
1. Compute dynamic threshold from current batch $A \leftarrow \varnothing$
Initialize list for all areas
for $\mathbf{y}_b \in \mathbf{y}_{batch}$ **do**
  $C_{list} \leftarrow$ Find all connected components in $\mathbf{y}_b$
  # Calculate area for each component
  for *each component c in* C_{list} **do**
    A.append(area(c))
$\tau \leftarrow$ Percentile(A, 90) # Batch-based threshold
2. Generate SO-GT mask $\mathbf{y}_s \leftarrow \varnothing$ # Initialize the Small-object GT batch
for $\mathbf{y}_b \in \mathbf{y}_{batch}$ **do**
  $\mathbf{y}_{small} \leftarrow$ CreateEmptyMask($\mathbf{y}_b$.shape)
  $C_{list} \leftarrow$ Find all connected components in $\mathbf{y}_b$
  for *each component c in* C_{list} **do**
    if *area*(c) $\le \tau$ **then**
      # Mark small object pixels
      $\mathbf{y}_{small}$[pixels(c)] $\leftarrow$ 255
  $\mathbf{y}_s$.append($\mathbf{y}_{small}$) # Add mask to the output batch
return $\mathbf{y}_s$

For the final prediction, all refined features are resized to the spatial resolution of $\tilde{\mathbf{P}}_1$, concatenated, compressed by a 3×3 convolution with batch normalization and GELU, upsampled once, and mapped by a 1×1 classifier to logits $\hat{\mathbf{y}}_m \in \mathbb{R}^{H\times W\times 2}$. To enhance the recall of subtle changes, the decoder implements our MS-SOE strategy. This strategy provides multiscale supervision via side heads attached to $\tilde{\mathbf{P}}_4$, $\tilde{\mathbf{P}}_3$, and $\tilde{\mathbf{P}}_2$,

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TABLE I
OVERVIEW OF THE FIVE BENCHMARK DATASETS. (T/V/T) DENOTES THE NUMBER OF PATCHES FOR TRAINING, VALIDATION, AND TESTING

Dataset	Study Site	Patch Pairs (T/V/T)	Size	Resolution	Evaluation Task
WHU-CD [3]	Christchurch, NZ	7,620 (6095 / 762 / 763)	256×256	0.3m	BCD / Semi-CD
LEVIR-CD [42]	Texas, USA	10,192 (7120 / 1024 / 2048)	256×256	0.5m	BCD / Semi-CD
CLCD [38]	Guangdong, China	2,400 (1440 / 480 / 480)	256×256	0.5–2.0m	BCD
CDD [43]	Google Earth	16,000 (10000 / 3000 / 3000)	256×256	0.03–1.0m	BCD
BANDON [44]	Various cities	2,283 (1689 / 202 / 392)	2048×2048	0.6m	Cross-Domain CD

which provide auxiliary supervision to improve convergence under class imbalance. In parallel, a lightweight small-object branch attached to $\tilde{\mathbf{P}}_1$ predicts $\hat{\mathbf{y}}_s$ under an area-thresholded mask $\mathbf{y}_s$. Instead of using a fixed mask, the small-object mask is generated during training according to the procedure described in Algorithm 1, as illustrated in the small-object ground-truth (SO-GT) panel of Fig. 9. For the training objective, we employ a loss composition that linearly combines Focal loss ($\mathcal{L}_f$) and Dice loss ($\mathcal{L}_d$) at each supervised head for stable optimization

$$rCl\mathcal{L}_{\text{fd}} = w_f\,\mathcal{L}_f + w_d\,\mathcal{L}_d. \tag{21}$$

The total loss $\mathcal{L}$ is a composite objective that combines the $\mathcal{L}_{\text{fd}}$ loss ($w_f = w_d = 0.5$) from the main head, the weighted side heads, and the small-object branch. For the multiscale deep supervision, the side-head predictions $\hat{\mathbf{y}}_k$, $k \in \{2,3,4\}$ are upsampled to the spatial resolution of the ground-truth $\mathbf{y}$ and weighted by $(\beta_2,\beta_3,\beta_4) = (0.5, 0.3, 0.1)$, prioritizing shallow, high-resolution features rich in spatial detail for precise boundary delineation [41]. For small-object enhancement, the small-object loss is restricted to the small-object mask $\mathbf{y}_s$, where the threshold is computed online as the 90th percentile of connected-component areas within each training batch. The overall objective is thus formulated as

$$\mathcal{L} = \mathcal{L}_{\text{fd}}\left(\hat{\mathbf{y}}_m, \mathbf{y}\right) + \sum_{k=2}^{4} \beta_k\,\mathcal{L}_{\text{fd}}\left(\hat{\mathbf{y}}_k, \mathbf{y}\right) + \mathcal{L}_{\text{fd}}\left(\hat{\mathbf{y}}_s, \mathbf{y}_s\right). \tag{22}$$

IV. EXPERIMENTS

A. Datasets

To comprehensively evaluate the effectiveness, robustness, and generalization capability of GCCANet, we conduct extensive experiments on five public benchmark datasets, including WHU-CD [3], LEVIR-CD [42], CLCD [38], change detection dataset (CDD) [43], and building change detection with off-nadir aerial images dataset (BANDON) [44]. The detailed statistics of these datasets are summarized in Table I. WHU-CD is derived from aerial images of Christchurch, New Zealand, and focuses on building changes in high-resolution urban scenes. LEVIR-CD contains large-scale high-resolution bi-temporal image pairs collected from multiple regions in Texas, USA, with annotations mainly related to building construction and demolition. CLCD is designed for cropland CD in Guangdong Province, China, where changed regions are often fragmented and land-cover patterns are relatively complex. CDD consists of season-varying remote sensing image pairs and introduces appearance variations caused by seasonal and illumination differences. BANDON focuses on building CD from off-nadir aerial images, where viewpoint-induced misalignment and facade ambiguity make CD more challenging.

B. Experimental Setup

1) Implementation Details: The method was implemented in PyTorch and trained on a single NVIDIA RTX 3090 graphics processing unit (GPU) with a batch size of 16. We used the AdamW optimizer with a weighted combination of focal and dice loss. Each input image is resized to $252 \times 252 \times 3$ pixels before training. This specific resolution is utilized to align with the patch size requirements of the DINOv2 backbone. The initial learning rates were set to 5×10^{-6} for the DINOv2 backbone, 1×10^{-4} for the ResNet backbone, and 2×10^{-4} for the remaining network components. A cosine annealing strategy was employed for learning rate scheduling, and the seed was 3407. After each training epoch, we conducted a test, and the model weights achieving the highest $F1$ score were saved as the best-performing model.

2) Evaluation Metrics: To compare our method's performance with SOTA approaches, we use five standard metrics for CD: Intersection over Union (IoU), $F1$-score ($F1$), Precision (Pre.), Recall (Rec.), and overall accuracy (OA). These metrics are formulated as follows:

$$\begin{cases} \text{Pre.} = \text{TP}/\left(\text{TP} + \text{FP}\right) \\ \text{Rec.} = \text{TP}/\left(\text{TP} + \text{FN}\right) \\ \text{F1} = \left(2 \times \text{Pre.} \times \text{Rec.}\right)/\left(\text{Pre.} + \text{Rec.}\right) \\ \text{IoU} = \text{TP}/\left(\text{TP} + \text{FN} + \text{FP}\right) \\ \text{OA} = \left(\text{TP} + \text{TN}\right)/\left(\text{TP} + \text{TN} + \text{FP} + \text{FN}\right). \end{cases} \tag{23}$$

C. Comparative Experiments

1) Quantitative Results: The methods we compare against are predominantly high-performing approaches from the last three years. Tables II and III report results on WHU-CD, LEVIR-CD, CLCD, and CDD, where GCCANet consistently achieves SOTA performance over recent CNN, Transformer, and Mamba baselines. On WHU-CD and LEVIR-CD, our model secures the top rank in key metrics like $F1$ and IoU, surpassing strong baselines such as Change3D and CDMamba. The performance gains are most pronounced on the challenging CLCD dataset, where GCCANet delivers a substantial improvement of 8.08 points in $F1$ and 11.12 points in IoU over the next best method. On CDD, our method again achieves the highest $F1$ and IoU scores. These notable improvements are closely related to the proposed semantic-guided and difference-aware design, especially in complex scenarios such as CLCD, which contains more fragmented changes,

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TABLE II

COMPARISON EXPERIMENTAL RESULTS ON WHU-CD AND LEVIR-CD DATASETS. THE UNITS FOR PARAMS ARE 10^6 (M) AND FLOPS ARE 10^9 (G). BEST VALUES ARE IN BOLD, SECOND BEST ARE UNDERLINED. * INDICATES RESULTS REPORTED FROM PREVIOUSLY PUBLISHED PAPERS

Method	Efficiency		WHU-CD					LEVIR-CD				
	Params	FLOPs	Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
FC-Siam-Diff* [14]	**1.35**	18.66	87.39	92.36	89.81	81.50	99.16	90.81	88.59	89.69	81.31	98.86
FC-Siam-Conc* [14]	1.54	21.07	86.57	91.11	88.78	79.83	99.08	91.41	88.43	89.89	81.64	98.98
ChangeViT [45]	32.13	38.80	95.08	93.87	94.47	89.52	99.56	92.72	90.88	91.79	84.83	99.17
HRMNet [46]	13.46	96.71	94.75	90.70	92.68	86.35	99.31	91.44	90.10	90.76	83.10	99.06
ScratchFormer [12]	36.99	196.59	94.76	89.91	92.28	85.66	99.26	92.25	89.98	91.10	83.66	99.13
ChangeMamba [29]	49.94	114.82	94.76	91.94	93.33	87.49	99.52	90.43	89.84	90.13	82.03	99.01
CDMamba [47]	10.99	259.49	95.58	92.01	93.76	88.26	99.51	91.43	90.08	90.75	83.07	99.06
DMINet [48]	6.24	**14.42**	94.89	92.02	93.43	87.68	99.48	92.16	88.83	90.46	82.59	99.04
Change3D [49]	1.66	15.19	95.65	**94.03**	94.83	90.17	99.21	92.01	91.14	91.58	84.46	99.06
GCCANet (Ours)	61.58	74.19	**96.64**	93.88	**95.24**	**90.91**	**99.64**	**93.10**	**91.40**	**92.22**	**85.56**	**99.21**

TABLE III

COMPARISON OF EXPERIMENTAL RESULTS ON CLCD AND CDD DATASETS. INFERENCE IS MEASURED IN MILLISECONDS (MS) AND FPS DENOTES FRAMES PER SECOND. BEST VALUES ARE IN BOLD, SECOND BEST ARE UNDERLINED. * INDICATES RESULTS REPORTED FROM PREVIOUSLY PUBLISHED PAPERS

Method	Efficiency		CLCD					CDD				
	Inference	FPS	Pre.	Rec.	F1	IoU	OA	Pre.	Rec.	F1	IoU	OA
FC-Siam-Diff* [14]	–	–	66.37	65.59	65.98	49.23	94.93	91.90	92.30	92.10	85.40	98.04
FC-Siam-Conc* [14]	–	–	71.30	62.45	66.58	49.91	95.33	94.00	95.40	94.70	89.90	98.67
ChangeViT [45]	29.41	34.01	82.42	69.26	75.27	60.35	96.61	97.52	97.72	97.62	95.35	99.41
HRMNet [46]	15.62	64.01	77.66	73.15	75.34	60.43	96.42	97.48	96.68	97.08	94.32	99.28
ScratchFormer [12]	96.14	10.40	69.50	55.53	61.73	44.65	94.72	96.68	96.65	96.66	93.54	99.17
ChangeMamba [29]	76.66	13.05	79.49	68.01	73.30	57.86	96.31	96.30	96.60	96.45	93.14	99.12
CDMamba [47]	62.44	16.02	67.08	61.03	63.91	46.96	94.87	95.29	95.07	95.18	90.81	98.81
DMINet [48]	**10.14**	**98.67**	73.24	70.11	71.64	55.81	95.87	96.68	96.46	96.57	93.37	99.15
Change3D [49]	20.28	49.31	80.20	70.71	75.16	60.20	96.39	97.12	97.72	97.42	94.97	99.36
GCCANet (Ours)	42.61	23.47	**87.34**	**79.83**	**83.42**	**71.55**	**97.64**	**97.54**	**98.14**	**97.84**	**95.77**	**99.49**

complex land-cover patterns, and stronger background interference, making pseudo-change suppression and fine-grained localization particularly challenging. By introducing multilevel semantic priors through ALA and BSM, GCCANet can better suppress background-induced pseudo-changes, while DLM explicitly models prechange, postchange, and difference features to strengthen discriminative bi-temporal representations. In addition, MS-SOE further enhances the perception of small and fragmented changed regions through multiscale supervision and a dedicated small-object branch. Fig. 10 further demonstrates the effectiveness of GCCANet in detecting small-object changes. From an efficiency perspective, it is important to acknowledge a limitation of GCCANet. As the tables indicate, its dual-backbone design results in the largest parameter count among the compared methods, representing a considerable memory footprint. Although its computational load floating-point operations (FLOPs) is not the lowest, it remains competitive and is notably lower than other complex methods like ChangeMamba, HRMNet, and ScratchFormer. GCCANet represents a deliberate tradeoff between accuracy and efficiency. While achieving superior detection performance, it maintains practical inference speed with an inference latency of 42.61 ms [23.47 frames per second (FPS)], outperforming recent Mamba-based models while remaining competitive with transformer-based approaches.

2) Qualitative Comparison: For qualitative comparison, representative scenarios from the four public datasets are presented in Figs. 11 and 12.

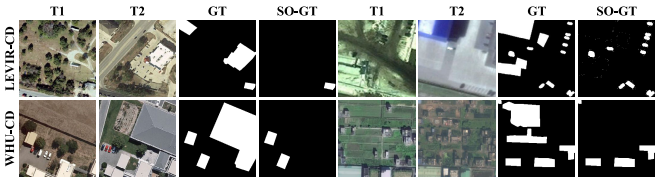


Fig. 9. Visualization of small-object segmentation labels in four datasets. T1 and T2 denote the prechange and postchange images, respectively, and SO-GT denotes the ground-truth change mask for small objects.

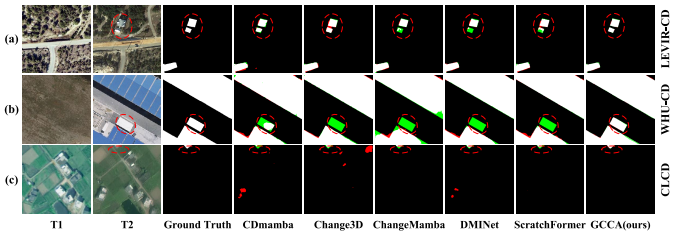


Fig. 10. Qualitative visualization of small-object CD across three benchmark datasets. White indicates true positive (TP), black true negative (TN), red FP, and green false negative (FN).

1) *WHU-CD and LEVIR-CD:* Both datasets present significant challenges in detecting building changes, particularly in preserving clear boundaries for large-scale or densely clustered structures and identifying small-scale changes. As shown in Fig. 11, GCCANet produces sharper and more continuous building boundaries with fewer breaks and holes, which benefits from the semantic guidance introduced by ALA and

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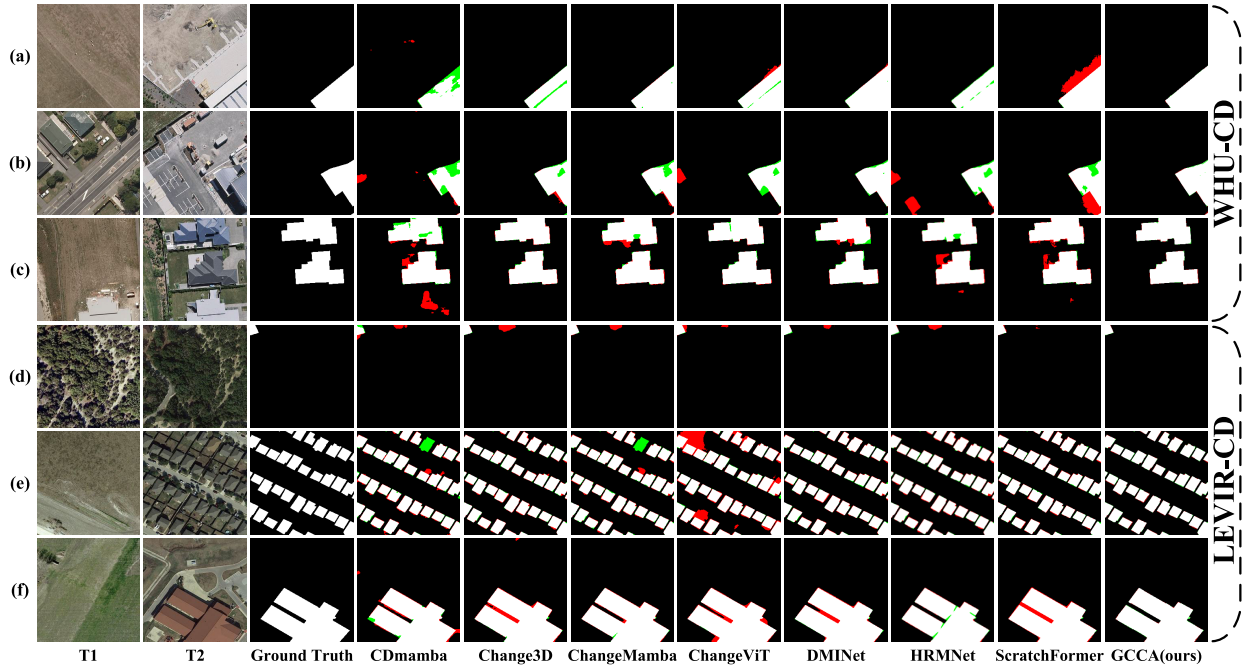


Fig. 11. Qualitative results on the WHU-CD and LEVIR-CD datasets. (a)–(c) Present representative examples from WHU-CD, while (d)–(f) show examples from LEVIR-CD. White denotes TPs, black TNs, red FPs, and green FNs.

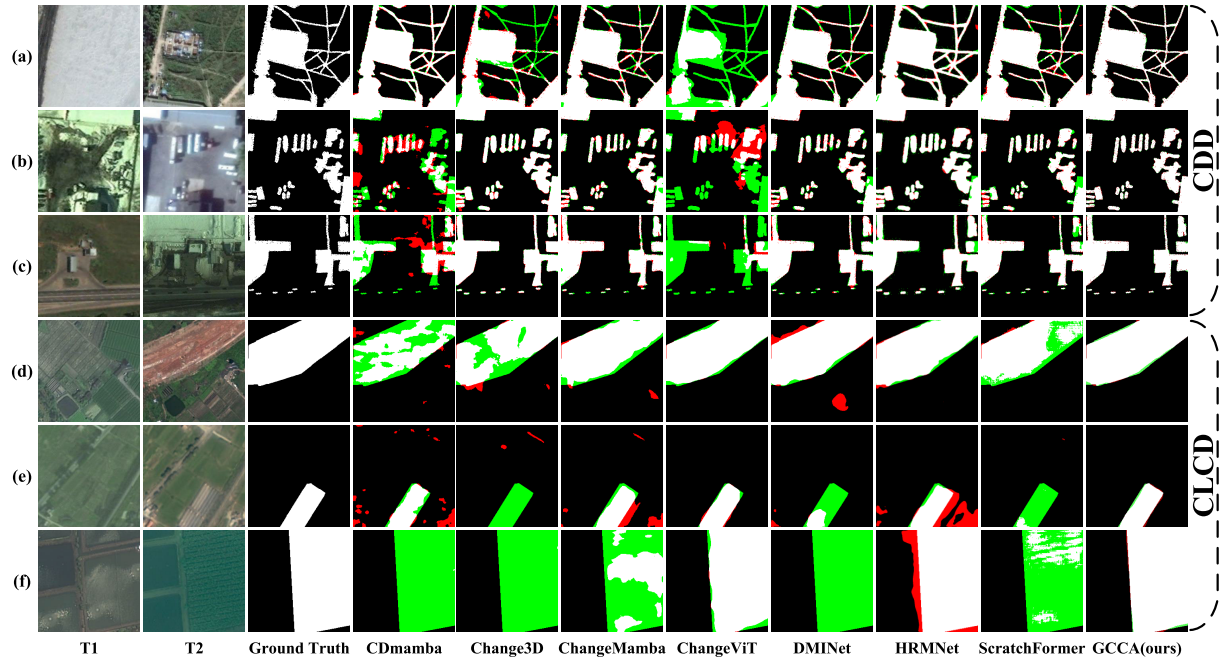


Fig. 12. Qualitative results on the CDD and CLCD datasets. (a)–(c) Present representative examples from CDD, while (d)–(f) show examples from CLCD. White denotes TPs, black TNs, red FPs, and green FNs.

BSM to suppress background responses around building contours. In Fig. 11(d), methods such as CDMamba and Change3D generate numerous FPs along building-change edges. Models like HRMNet and ScratchFormer exhibit internal holes and missed detections within changed buildings, whereas GCCANet effectively avoids these false responses and omissions by jointly modeling prechange, postchange, and difference features through DLM. In Fig. 11(e), ChangeViT and ChangeMamba show both missed detections and FPs in dense building regions. Our method produces cleaner and more complete CMaps. As shown in Fig. 11(f), while most methods blur or break object boundaries, our approach preserves sharp edges and achieves nearly flawless results.

2) *CLCD and CDD:* Both datasets pose challenges for land-cover CD due to their mixed building and surface changes, unclear boundaries, and pseudo-changes arising from complex surface dynamics and illumination variations. As shown in Fig. 12, on the CLCD dataset, our method delineates parcel boundaries and agricultural patterns with minimal edge erosion, indicating

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TABLE IV

SMALL-OBJECT CD RESULTS ON THREE DATASETS. BEST VALUES ARE IN BOLD, AND THE SECOND BEST ARE UNDERLINED

Method	WHU-CD			LEVIR-CD			CLCD		
	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
CDMamba [47]	79.87	66.49	99.13	74.40	59.24	98.34	40.70	25.55	97.07
ScratchFormer [12]	80.79	67.77	99.19	85.82	75.16	99.00	36.91	22.63	97.49
ChangeMamba [29]	84.23	72.76	99.33	82.32	69.95	98.79	48.10	31.67	97.65
DMINet [48]	87.16	77.24	99.45	85.66	74.91	98.99	40.55	25.43	97.46
Change3D [49]	88.16	78.82	99.49	85.68	74.95	98.99	46.17	30.01	97.09
GCCANet(Ours)	**91.43**	**84.21**	**99.61**	**87.75**	**78.18**	**99.11**	**61.06**	**43.95**	**98.21**

that the proposed semantic-guided design can better stabilize local responses under complex land-cover backgrounds. In contrast, most competing approaches fail to identify the large-scale changed region within the agricultural field. As shown in Fig. 12(a), most baselines produce fragmented outputs along complex linear structures, whereas our method preserves their continuity and completeness. In Fig. 12(b), CDMamba and ChangeViT exhibit missed and false detections when identifying small objects, while our approach detects all of them, benefiting from MS-SOE, which enhances the perception of small and fragmented changed regions through multiscale supervision and a dedicated small-object branch.

D. Small-Object CD

Identifying small-scale and subtle changes remains a long-standing challenge in RSCD, as fine-grained structures are easily overwhelmed by background noise or degraded during feature downsampling and upsampling. To further assess GCCANet's capability in small-object CD, we adopt a dedicated evaluation protocol where the prediction maps produced by all methods are compared against ground-truth masks containing only small-object changes. The quantitative results in Table IV, together with the visual comparisons in Fig. 10, show that GCCANet achieves consistently superior performance. As illustrated in Fig. 10(a), most methods detect changes in the lower, well-illuminated building but fail to capture the two small buildings affected by illumination, whereas GCCANet identifies them accurately. In Fig. 10(b), competing approaches largely miss the small-object changes, while our model produces stable detections. For the CLCD sample with seasonal variations [Fig. 10(c)], existing methods overlook subtle field changes, but GCCANet successfully captures these fine-grained differences. Across all datasets, GCCANet achieves the highest $F1$ and IoU scores, with particularly notable gains on the challenging CLCD dataset, where the $F1$-score reaches 61.06%, substantially surpassing the second-best method.

E. Semi-Supervised CD

Label efficiency is a critical constraint in practical RSCD. To evaluate the robustness and label dependence of GCCANet, we conducted semi-supervised experiments on the WHU-CD and LEVIR-CD datasets using 5%–40% labeled data, adopting a consistency-based framework inspired by UniMatch [53]. The results in Tables V and VI show that GCCANet

TABLE V

ACCURACY VERSUS LABELED-DATA RATIOS ON THE WHU-CD DATASET. BEST VALUES ARE IN BOLD, AND THE SECOND BEST ARE UNDERLINED

Method	5%	10%	20%	40%
S4GAN [50]	18.3 / 96.7	62.6 / 98.2	70.8 / 98.6	76.4 / 98.9
SemiCDNet [51]	51.7 / 97.7	68.1 / 98.3	76.1 / 98.6	79.2 / 98.9
SemiCD [52]	65.8 / 98.4	71.5 / 98.5	77.0 / 98.7	82.5 / 98.9
UniMatch [53]	**80.2 / 99.2**	**81.7 / 99.2**	81.7 / 99.2	85.1 / 99.4
SAM-CD [22]	66.9 / 98.6	71.4 / 98.7	77.9 / 99.0	84.0 / 99.2
Ours (Label Only)	73.5 / 98.9	76.7 / 98.9	83.3 / 99.3	86.1 / 99.4
Ours (SSL)	70.1 / 98.6	78.6 / 99.1	**84.8 / 99.3**	**87.6 / 99.5**

TABLE VI

ACCURACY VERSUS LABELED-DATA RATIOS ON THE LEVIR-CD DATASET. BEST VALUES ARE IN BOLD, AND THE SECOND BEST ARE UNDERLINED

Method	5%	10%	20%	40%
S4GAN [50]	64.0 / 97.9	67.0 / 98.1	73.4 / 98.5	75.4 / 98.6
SemiCDNet [51]	67.6 / 98.2	71.5 / 98.4	74.3 / 98.6	75.4 / 98.6
SemiCD [52]	72.5 / 98.5	75.5 / 98.6	76.2 / 98.7	77.2 / 98.7
SAM-CD [22]	73.9 / 98.5	79.2 / 98.9	80.2 / 98.9	83.2 / 99.1
UniMatch [53]	80.7 / 98.9	81.5 / 98.9	81.7 / 99.0	82.1 / 99.0
Ours (Label Only)	**81.9 / 99.0**	82.0 / 99.0	82.9 / 99.0	82.9 / 99.0
Ours (SSL)	81.2 / 98.9	**83.0 / 99.0**	**83.0 / 99.0**	**83.4 / 99.1**

demonstrates strong label efficiency, as it achieves competitive performance even in a supervised-only setting with very limited annotations (Label Only). On LEVIR-CD, the supervised-only model reaches an IoU of 81.9% with only 5% labeled data, which already exceeds most competing semi-supervised methods, such as SemiCD (72.5% IoU). A similar trend is observed on WHU-CD, where our supervised-only baseline attains IoU scores of 73.5% and 76.7% using 5% and 10% labeled data, respectively. The consistency-regularized semi-supervised learning (SSL) strategy further improves performance at 10%, 20%, and 40% label ratios on both datasets. A slight performance drop occurs at the 5% label ratio, likely because our architecture already produces highly stable predictions, making consistency-based regularization less effective when noisy pseudo-labels dominate the unsupervised portion of the training data.

F. Cross-Domain CD

Cross-domain CD is a key issue for practical deployment, as annotating specialized data for each scene is costly. Therefore, a model is expected to perform well not only on in-domain data but also on out-of-domain data, demonstrating strong generalization ability. The BANDON dataset is widely used as a cross-domain benchmark in RSCD [56], [57]. It provides explicit in-domain and out-of-domain splits from different cities, enabling direct evaluation of cross-domain generalization. Moreover, it covers diverse urban and rural regions captured under different sensors, illumination conditions, and seasonal variations, creating a challenging setting for domain generalization. The component-wise ablation results are provided in Table VII for complementary analysis. To assess this for GCCANet, we validated it on the out-of-domain test split of the BANDON dataset. As shown in Table VIII, cross-domain inference leads to clear degradation for existing methods; DTCDSCN's IoU drops from 46.05% to 38.43%, and Change-Mix's IoU drops from 53.89% to

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TABLE VII

ABLATION STUDIES ON THREE PUBLIC DATASETS. THE UNITS FOR PARAMS ARE 10^6 (M), FLOPS ARE 10^9 (G). ACCURACY METRICS ARE EXPRESSED IN PERCENTAGES (%)

No.	Model	Efficiency		LEVIR-CD			WHU-CD			CLCD		
		Params	FLOPs	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU
#01	GCCANet	61.58	74.19	99.21	92.22	85.56	99.64	95.24	90.91	97.64	83.42	71.55
#02	w/o ALA	61.55	74.18	99.20	92.12	85.39	99.62	95.07	90.61	97.52	82.94	70.85
#03	w/o BSM	54.68	70.17	99.19	91.98	85.15	99.61	95.02	90.51	96.99	79.60	66.11
#04	w/o DLM	55.64	70.45	99.17	91.83	84.89	99.59	94.83	90.18	97.01	77.48	63.24
#05	w/o global spatial attention	61.32	73.18	99.20	92.09	85.34	99.62	95.22	90.88	97.45	82.34	69.98
#06	w/o MS-SOE multi-scale branch	61.57	74.18	99.19	91.92	85.05	99.59	94.72	89.97	97.30	81.91	69.37
#07	w/o MS-SOE small object branch	61.57	74.18	99.20	92.03	85.24	99.61	95.04	90.55	97.46	82.28	69.90

TABLE VIII

CROSS-DOMAIN EXPERIMENTS ON THE BANDON DATASET. BEST VALUES ARE IN BOLD, AND THE SECOND BEST ARE UNDERLINED

Method	In-domain Test				Out-of-domain Test			
	Pre.	Rec.	F1	IoU	Pre.	Rec.	F1	IoU
FC-EF [14]	51.80	34.83	41.65	23.30	40.76	30.76	35.04	21.24
FC-Siam-Diff [14]	65.11	67.39	66.23	49.51	60.33	60.06	60.20	43.06
FC-Siam-Conc [14]	65.49	66.17	65.83	49.06	60.81	58.65	59.71	42.56
DTCDSCN [54]	69.23	57.90	63.06	46.05	54.22	56.89	55.53	38.43
STANet [42]	66.24	69.42	67.79	51.28	60.35	61.92	61.12	44.01
Change-Mix [55]	69.29	**70.80**	70.04	53.89	65.60	63.26	64.41	47.50
GCCANet (Ours)	**71.48**	70.59	**71.03**	**55.08**	**67.60**	**63.66**	**65.57**	**48.78**

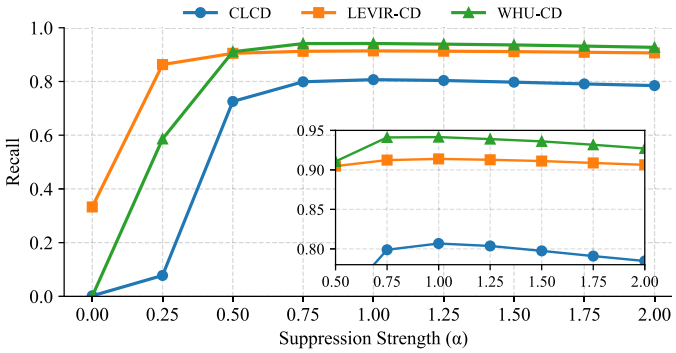


Fig. 13. Recall sensitivity to the suppression strength α across three datasets. The inset highlights the slight decrease when $\alpha > 1$.

47.50%. GCCANet attains an IoU of 55.08% and an $F1$ of 71.03% in-domain, and an IoU of 48.78% and an $F1$ of 65.57% out-of-domain. On the out-of-domain test set, GCCANet surpasses Change-Mix, one of the best methods, by 1.28 points in IoU and 1.16 points in $F1$. Although performance degrades compared to the in-domain results, this degradation is effectively mitigated, indicating that GCCANet possesses strong domain generalization capability.

G. Ablation Study

Six ablation configurations and one analysis were conducted on three public datasets to quantify the effect of each component and training strategy in GCCANet. The module-wise results are given in Table VII, where #01 denotes the full model. The sensitivity to the suppression strength and the qualitative background suppression effect are further illustrated in Figs. 13 and 14. To provide a more intuitive validation, we also visualize the internal feature heatmaps for key modules (Fig. 15).

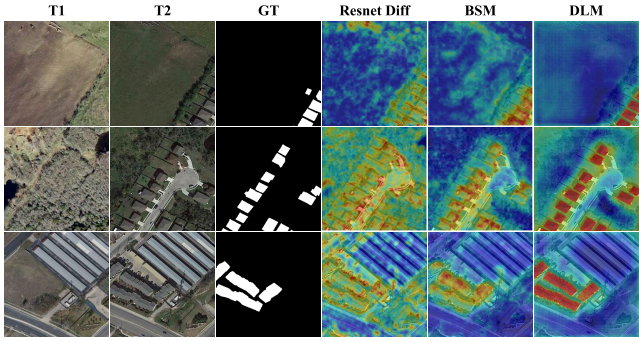


Fig. 14. Visualization of the background suppression effect of BSM on three LEVIR-CD test samples, where ResNet Diff is computed as the absolute difference between the bi-temporal ResNet features.

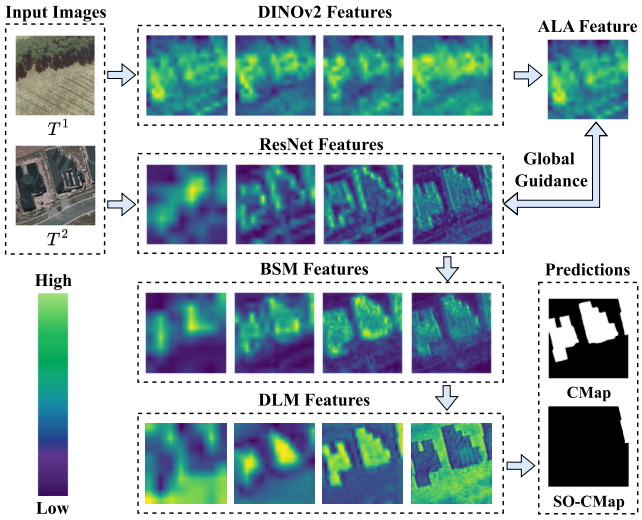


Fig. 15. Visualization of the feature refinement process on a WHU-CD test sample. For each feature tensor of size $B \times C \times H \times W$, a response map is obtained by averaging across the channel dimension of one sample in the batch and visualized using a color map with a color bar. The figure illustrates the progression from Backbone (DINOv2, ResNet) features and ALA guidance, through BSM refinement and DLM difference encoding, to the prediction CMap and small-object CMap (SO-CMap).

1) Effectiveness of ALA: As shown in experiment #02 of Table VII, we replaced the ALA module's dynamic aggregation with a variant that uses only the feature map from the final DINOv2 layer as the global guidance. This replacement resulted in $F1$ and IoU scores decreasing by 0.10% and 0.17% on LEVIR-CD, by 0.17% and 0.30% on WHU-CD, and by 0.48% and 0.70% on CLCD. This confirms that our approach

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TABLE IX
EFFECT OF DIFFERENT SMALL-OBJECT THRESHOLDS

Dataset	70%		80%		90%	
	Rec.	F1	Rec.	F1	Rec.	F1
CLCD	80.37	82.48	79.87	82.17	79.83	83.42
LEVIR-CD	91.67	92.22	91.55	92.09	91.40	92.22
WHU-CD	93.96	95.16	93.87	95.08	93.88	95.24

TABLE X
SENSITIVITY ANALYSIS OF FOCAL AND DICE LOSS WEIGHTS

Dataset	Focal	Dice	Pre.	Rec.	F1	IoU	OA
CLCD	1.00	0.00	85.93	79.81	82.75	70.58	97.53
	0.25	0.75	86.72	79.70	83.06	71.03	97.58
	0.75	0.25	87.40	79.27	83.14	71.14	97.61
	0.00	1.00	86.33	79.42	82.73	70.54	97.53
	0.50	**0.50**	**87.34**	**79.83**	**83.42**	**71.55**	**97.64**
LEVIR-CD	1.00	0.00	92.67	91.38	92.02	85.21	99.19
	0.25	0.75	92.98	91.32	92.14	85.43	99.21
	0.75	0.25	92.56	91.58	92.07	85.31	99.20
	0.00	1.00	92.23	91.83	92.03	85.23	99.19
	0.50	**0.50**	**93.10**	**91.40**	**92.22**	**85.56**	**99.21**
WHU-CD	1.00	0.00	95.86	94.15	95.00	90.48	99.61
	0.25	0.75	96.17	94.12	95.13	90.72	99.62
	0.75	0.25	96.00	94.27	95.13	90.71	99.62
	0.00	1.00	95.99	94.05	95.01	90.49	99.61
	0.50	**0.50**	**96.64**	**93.88**	**95.24**	**90.91**	**99.64**

TABLE XI
STABILITY ANALYSIS UNDER DIFFERENT RANDOM SEEDS

Seed	Dataset	Pre.	Rec.	F1	IoU
2024	CLCD	85.99	78.82	82.25	69.85
	LEVIR-CD	93.01	91.27	92.13	85.41
	WHU-CD	96.40	93.96	95.16	90.77
2025	CLCD	86.87	79.14	82.83	70.69
	LEVIR-CD	92.53	91.86	92.19	85.51
	WHU-CD	96.44	94.15	95.28	90.99
2026	CLCD	86.36	80.22	83.18	71.20
	LEVIR-CD	92.98	91.37	92.17	85.47
	WHU-CD	96.19	94.27	95.22	90.88
3407	CLCD	87.34	79.83	83.42	71.55
	LEVIR-CD	93.10	91.40	92.22	85.56
	WHU-CD	96.64	93.88	95.24	90.91

of aggregating multilevel semantics is more effective than relying solely on the final layer.

2) Effectiveness of BSM and Sensitivity of the Suppression Strength: To evaluate the effectiveness of the BSM, experiment #03 in Table VII replaces its guided fusion mechanism (implemented with ERFB) with a naive element-wise summation of the DINOv2 and ResNet features. This modification leads to decreases of 0.24% and 0.41% in $F1$ and IoU on LEVIR-CD, 0.22% and 0.40% on WHU-CD, and a larger drop of 3.82% and 5.44% on CLCD, indicating that incorporating background priors is particularly beneficial in cluttered and low-contrast scenarios. Qualitative visualization in Fig. 14 further compares the difference responses obtained from CNN-only features (ResNet Diff) and from BSM-refined features, where the refined features enable the subsequent DLM to produce clearer and more stable structural responses. We further vary the injection coefficient α in the BSM to analyze the effect of background suppression strength. As shown in Fig. 13, when $\alpha = 0$, the global guidance is disabled, resulting in low Recall across all datasets. As α increases, Recall improves significantly and reaches a stable optimum near 1.0. When $\alpha > 1$, overly strong suppression begins to attenuate useful change signals, causing a slight decrease in Recall.

3) Effectiveness of DLM and Its GSA Component: To evaluate the DLM, experiment #04 of Table VII replaced the entire module with a simple baseline that computes the absolute difference of the features followed by a 1×1 convolution. This variant exhibits the most severe degradation among all ablations. $F1$ and IoU scores decreased by 0.39% and 0.67% on LEVIR-CD, by 0.41% and 0.73% on WHU-CD, and substantially by 5.94% and 8.31% on CLCD. We further isolated the GSA mechanism in experiment #05, where this internal block was replaced with an identity mapping. This also caused a notable performance drop. $F1$ and IoU scores dropped by 0.13% and 0.22% on LEVIR-CD, by 0.02% and 0.03% on WHU-CD, and by 1.08% and 1.57% on CLCD. These two sets of results confirm that structured difference modeling is essential for capturing reliable bi-temporal changes, and that GSA effectively integrates local details and global context, thereby strengthening the representation capability of DLM.

4) Effectiveness of Ms-SOE: To evaluate the MS-SOE strategy, we independently removed its two components. As shown in experiment #06 of Table VII, removing the multiscale side heads resulted in consistent performance drops across all datasets, with the largest decreases on CLCD. Experiment #07 shows that removing the small-object head also leads to noticeable degradation. These results suggest that the multiscale side heads strengthen regional consistency through deep supervision, while the small-object head helps preserve fine boundaries and microscale changes.

5) Effect of the Small-Object Threshold: To evaluate the sensitivity of the small-object threshold, we conducted additional experiments with different percentile values (70%, 80%, and 90%). As shown in Table IX, the performance differences across thresholds remain very small on all datasets in terms of both Recall and $F1$. The slightly larger Recall variation observed on CLCD results from its fragmented cropland changes and irregular object sizes, which make the percentile-based small-object definition more sensitive to the threshold.

6) Effectiveness of Focal and Dice Loss Weight Configuration: To examine the influence of loss-weight configurations, we conducted additional experiments with different combinations of Focal loss and Dice loss. As reported in Table X, the equal-weight setting achieves the most favorable overall performance on CLCD, LEVIR-CD, and WHU-CD, particularly in terms of $F1$ and IoU, indicating that a balanced combination of the two losses provides a reasonable tradeoff between pixel-wise hard-sample learning and region-level overlap optimization. In contrast, weight configurations biased toward either Focal loss or Dice loss do not consistently improve the comprehensive performance across datasets. Moreover, the performance differences among different loss-weight settings remain relatively small, with only marginal variations in the main evaluation metrics, indicating that the loss-weight configuration has a limited effect on the final performance. The effectiveness of GCCANet mainly originates from its

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semantic-guided background suppression, structured difference modeling, and small-object enhancement designs, rather than from a particular choice of Focal and Dice loss weights.

7) Robustness to Different Random Seeds: To further evaluate the training stability of GCCANet, we conducted experiments under different random seeds. As shown in Table XI, GCCANet maintains stable performance across different seeds on all three datasets. The $F1$ fluctuation is only 0.09% on LEVIR-CD and 0.12% on WHU-CD, while the variation on CLCD is slightly larger at 1.17%. This larger fluctuation is likely related to the more complex cropland scenes in CLCD, where changed regions are more fragmented and background interference is stronger. Nevertheless, GCCANet consistently maintains high $F1$ and IoU scores under different seeds, demonstrating that the performance gains are not caused by random initialization but mainly come from the proposed semantic-guided background suppression, structured difference modeling, and small-object enhancement designs.

V. Conclusion

In this article, we presented GCCANet, a dual-backbone architecture that synergizes a Transformer for global semantic context with a CNN for fine-grained local details. The core of our method lies in a novel guided fusion mechanism, where global priors are used to modulate local features, and in an advanced DLM that enhances change discrimination. Paired with a deeply supervised multiscale decoder, GCCANet achieves SOTA performance on five public benchmarks, showing particular strengths in boundary integrity and small-object recall. Extensive ablation studies systematically validated the contributions of our key architectural components. While the dual-backbone design yields high accuracy, its primary limitations are the considerable computational cost and the reliance on a percentile-based heuristic for small-object supervision. Future work will focus on developing more parameter-efficient variants, extending the framework to semantic CD, and exploring more principled small-object supervision strategies.

References

[1] X. Li, F. Ling, G. M. Foody, and Y. Du, "A superresolution land-cover change detection method using remotely sensed images with different spatial resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 7, pp. 3822–3841, Jul. 2016.

[2] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "Building damage assessment for rapid disaster response with a deep object-based semantic change detection framework: From natural disasters to man-made disasters," *Remote Sens. Environ.*, vol. 265, Nov. 2021, Art. no. 112636.

[3] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[4] C.-F. Chen et al., "Multi-decadal mangrove forest change detection and prediction in honduras, central america, with Landsat imagery and a Markov chain model," *Remote Sens.*, vol. 5, no. 12, pp. 6408–6426, Nov. 2013.

[5] S. H. Khan, X. He, F. Porikli, and M. Bennamoun, "Forest change detection in incomplete satellite images with deep neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5407–5423, Sep. 2017.

[6] W. Hua and Q. Chen, "A survey of small object detection based on deep learning in aerial images," *Artif. Intell. Rev.*, vol. 58, no. 6, pp. 1–67, Mar. 2025.

[7] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning-based change detection in remote sensing images: A review," *Remote Sens.*, vol. 14, no. 4, p. 871, Feb. 2022.

[8] X. Dai and S. Khorram, "Quantification of the impact of misregistration on the accuracy of remotely sensed change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. Remote Sens.-Sci. Vis. Sustain. Develop. (IGARSS)*, vol. 4, Aug. 1997, pp. 1763–1765.

[9] S. Marchesi and L. Bruzzone, "ICA and kernel ICA for change detection in multispectral remote sensing images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2009, pp. II-980–II-983.

[10] D. K. Seo, Y. H. Kim, Y. D. Eo, M. H. Lee, and W. Y. Park, "Fusion of SAR and multispectral images using random forest regression for change detection," *ISPRS Int. J. Geo-Inf.*, vol. 7, no. 10, p. 401, Oct. 2018.

[11] H. Zhang, M. Lin, G. Yang, and L. Zhang, "ESCNet: An end-to-end superpixel-enhanced change detection network for very-high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 1, pp. 28–42, Jan. 2023.

[12] M. Noman et al., "Remote sensing change detection with transformers trained from scratch," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4704214.

[13] L. Mou, L. Bruzzone, and X. X. Zhu, "Learning spectral–spatial–temporal features via a recurrent convolutional neural network for change detection in multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 924–935, Feb. 2019.

[14] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[15] T. Liu et al., "Building change detection for VHR remote sensing images via local–global pyramid network and cross-task transfer learning strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–17, 2022.

[16] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[17] M. Liu, Q. Shi, A. Marinoni, D. He, X. Liu, and L. Zhang, "Super-resolution-based change detection network with stacked attention module for images with different resolutions," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–18, 2022.

[18] L. Yan and J. Jiang, "A hybrid Siamese network with spatiotemporal enhancement and two-level feature fusion for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4403217.

[19] V. Ashish, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–11.

[20] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[21] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[22] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.

[23] L. Yang, Z. Zhao, and H. Zhao, "UniMatch v2: Pushing the limit of semi-supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 3031–3048, Apr. 2025.

[24] P. F. Alcantarilla, S. Stent, G. Ros, R. Arroyo, and R. Gherardi, "Street-view change detection with deconvolutional networks," *Auto. Robots*, vol. 42, no. 7, pp. 1301–1322, Oct. 2018.

[25] R. Shao, C. Du, H. Chen, and J. Li, "SUNet: Change detection for heterogeneous remote sensing images from satellite and UAV using a dual-channel fully convolution network," *Remote Sens.*, vol. 13, no. 18, p. 3750, Sep. 2021.

[26] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[27] A. Gu, K. Goel, and C. Ré, "Efficiently modeling long sequences with structured state spaces," 2021, *arXiv:2111.00396*.

[28] J. T. Smith, A. Warrington, and S. Linderman, "Simplified state space layers for sequence modeling," in *Proc. 11th Int. Conf. Learn. Represent.*, 2023, pp. 1–35.

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[29] H. Chen, J. Song, C. Han, J. Xia, and N. Yokoya, "ChangeMamba: Remote sensing change detection with spatiotemporal state space model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4409720.

[30] S. Zhao, H. Chen, X. Zhang, P. Xiao, L. Bai, and W. Ouyang, "RS-mamba for large remote sensing image dense prediction," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5633314.

[31] D. Duan, P. Liu, B. Hui, and F. Wen, "Brain-inspired online adaptation for remote sensing with spiking neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5605818.

[32] Y. Pang, L. Yao, Y. Luo, C. Dong, Q. Kong, and B. Chen, "RepSViT: An efficient vision transformer based on spiking neural networks for object recognition in satellite on-orbit remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610916.

[33] S. Yang, B. Linares-Barranco, Y. Wu, and B. Chen, "Self-supervised high-order information bottleneck learning of spiking neural network for robust event-based optical flow estimation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 4, pp. 2280–2297, Apr. 2025.

[34] J. Seo, W. Park, and T. Kim, "Feature-based approach to change detection of small objects from high-resolution satellite images," *Remote Sens.*, vol. 14, no. 3, p. 462, Jan. 2022.

[35] B. Yang, Y. Huang, X. Su, and H. Guo, "MAEANet: Multiscale attention and edge-aware Siamese network for building change detection in high-resolution remote sensing images," *Remote Sens.*, vol. 14, no. 19, p. 4895, Sep. 2022.

[36] X. Ma et al., "STNet: Spatial and temporal feature fusion network for change detection in remote sensing images," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 2195–2200.

[37] M. Liu and Q. Shi, "DSAMNet: A deeply supervised attention metric based network for change detection of high-resolution images," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 6159–6162.

[38] M. Liu, Z. Chai, H. Deng, and R. Liu, "A CNN-transformer network with multiscale context aggregation for fine-grained cropland change detection," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 4297–4306, 2022.

[39] B. Koonce, "Resnet 50," in *Convolutional Neural Networks With Swift for Tensorflow: Image Recognition and Dataset Categorization*. Cham, Switzerland: Springer, 2021, pp. 63–72.

[40] R. Xiong et al., "On layer normalization in the transformer architecture," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Vienna, Austria, Jul. 2020, pp. 10524–10533.

[41] C.-Y. Lee, S. Xie, P. Gallagher, Z. Zhang, and Z. Tu, "Deeply-supervised nets," in *Proc. 18th Int. Conf. Artif. Intell. Statist.*, vol. 38, 2015, pp. 562–570.

[42] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[43] M. Lebedev, Y. V. Vizilter, O. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 42, pp. 565–571, May 2018.

[44] C. Pang, J. Wu, J. Ding, C. Song, and G.-S. Xia, "Detecting building changes with off-nadir aerial images," *Sci. China Inf. Sci.*, vol. 66, no. 4, Apr. 2023, Art. no. 140306.

[45] D. Zhu, X. Huang, H. Huang, Q. Cheng, Z. Huang, and Z. Shao, "ChangeViT: Unleashing plain vision transformers for change detection in remote sensing images," *Pattern Recognit.*, vol. 172, Apr. 2026, Art. no. 112539.

[46] Z. Li et al., "Boosting remote sensing change detection via hard region mining," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5639812.

[47] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[48] Y. Feng, J. Jiang, H. Xu, and J. Zheng, "Change detection on remote sensing images using dual-branch multilevel intertemporal network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4401015.

[49] D. Zhu, X. Huang, H. Huang, H. Zhou, and Z. Shao, "Change3D: Revisiting change detection and captioning from a video modeling perspective," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2025, pp. 24011–24022.

[50] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2021.

[51] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[52] W. Gedara Chaminda Bandara and V. M. Patel, "Revisiting consistency regularization for semi-supervised change detection in remote sensing images," 2022, *arXiv:2204.08454*.

[53] L. Yang, L. Qi, L. Feng, W. Zhang, and Y. Shi, "Revisiting weak-to-strong consistency in semi-supervised semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 7236–7246.

[54] Y. Liu, C. Pang, Z. Zhan, X. Zhang, and X. Yang, "Building change detection for remote sensing images using a dual-task constrained deep Siamese convolutional network model," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 5, pp. 811–815, May 2021.

[55] Z. Zheng, A. Ma, L. Zhang, and Y. Zhong, "Change is everywhere: Single-temporal supervised object change detection in remote sensing imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15193–15202.

[56] K. Li, X. Cao, and D. Meng, "A new learning paradigm for foundation model-based remote-sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5610112.

[57] M. Esfandiari, G. Abdi, S. Jabari, and V. S. P. Lolla, "Building change detection in off-nadir images using deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 1347–1350.

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